ABHIGYAN CHAKRABORTY

Current Position: PhD Student, Department of Civil Engineering, IIT Hyderabad
Research Area: Aerosol-Cloud-Precipitation Interactions
Email: ce23resch12001@iith.ac.in
Website: Abhigyan-pro.github.io

Research Summary

Aerosols can either enhance or suppress precipitation, and changes in aerosol pollution can alter precipitation and its extremes in a warming climate. My research focuses on understanding these aerosol–cloud–precipitation interactions using long-term observations from satellites, reanalysis, and in-situ measurements. I use statistical and unsupervised machine learning to disentangle the influence of aerosols on precipitation from variability in the surrounding meteorological conditions. I also use numerical simulations to test hypotheses and investigate the underlying physical processes, while exploring surrogate modeling and digital twins for weather and climate research.

Research & Technical Skills:

- Satellite Remote Sensing
- Machine Learning
- High Performance Computing

Education

Sep 2023–Present	**Indian Institute of Technology Hyderabad** Doctor of Philosophy in Civil Engineering	CGPA 9.5/10
Jul 2020–May 2022	**National Institute of Technology Agartala** Master of Technology in Civil Engineering (specializing in Environmental Engineering) Masters Thesis: Development and validation of Machine learning predictive models for the estimation of soil moisture content using IoT-based field data, remote sensing derivatives, and satellite imagery.	CGPA 8.91/10
Jul 2015– May 2019	**KIIT University** Bachelor of Technology in Civil Engineering	CGPA 8.25/10

Professional Experience

Sep 2022 – Sep 2023

North Eastern Space Application Centre (NESAC)
Department of Space, Govt. of India

Position : Junior Research Fellow

Division : Water Resources

Project : Flood Early Warning system for Northeast India & Flood Risk Map development for Assam State, India

Publications:

Peer-Reviewed Journal Articles

Mallik, S., **Chakraborty, A.**, Podder, K., Talukdar, S., Rahman, A., & Mishra, U. (2025). Enhancing Soil Moisture Prediction with Explainable AI: Integrating IoT and Multi-Sensor Remote Sensing Data through Soft Computing. Applied Soft Computing.[DOI]

Mallik, S., Das, S., **Chakraborty, A.**, Mishra, U., Talukdar, S., Bera, S., & Ramana, G. V. (2023). Prediction of non-carcinogenic health risk using Hybrid Monte Carlo-machine learning approach. Human and Ecological Risk Assessment. [DOI]

Mallik, S., **Chakraborty, A.**, Mishra, U., & Paul, N. (2022). Prediction of irrigation water suitability using geospatial computing approach: a case study of Agartala city, India. Environmental Science and Pollution Research. [DOI]

Peer-Reviewed Conference Proceedings

Chakraborty, A., Arulraj, M., & Upadhyaya, S. (2026). Performance Assessment of GPM IMERG Products Using a Multi-Triplet Collocation Approach. In 2026 IEEE International Geoscience and Remote Sensing Symposium. **(Accepted)**

Chakraborty, A., Pol, U., & Upadhyaya, S. (2026). Observational Evidence of Microphysical and Structural Variability in Monsoon Precipitation Under Contrasting Aerosol Conditions. In 2026 IEEE India Geoscience and Remote Sensing Symposium (InGARSS). **(Accepted)**

Kumar, B., **Chakraborty, A.**, Upase, Y., Khadagaraj, S., Das, S. & Upadhyaya, S. (2026). Can Spatial Context from Satellite Observations Improve Local Rain Gauge Precipitation Nowcasting? In 2026 IEEE India Geoscience and Remote Sensing Symposium (InGARSS). **(Accepted)**

Baghel, A., Jha, U., Upase, Y., **Chakraborty, A.**, & Upadhyaya, S. (2025, December). Flash Drought Forecasting for Central India Using Long Short-Term Memory. In 2025 IEEE India Geoscience and Remote Sensing Symposium (InGARSS). IEEE. [DOI]

Chakraborty, A., Kumar, B., & Upadhyaya, S. (2024, December). Predicting and Mapping Flood Susceptibility: Leveraging Explainable AI and GIS Techniques. In 2024 IEEE India Geoscience and Remote Sensing Symposium (InGARSS). IEEE. [DOI]

Atheeswaran, B., Jain, S., **Chakraborty, A.**, & Upadhyaya, S. (2024, December). Enhancing River Discharge Forecasting With Multivariate LSTM. In 2024 IEEE India Geoscience and Remote Sensing Symposium (InGARSS). IEEE. [DOI]

Under Review / Submitted

Chakraborty, A., Upase, Y., Upadhyaya, S., Arulraj, M., & Emmanuel-Kirstetter, P. (2025). Performance of GSMaP-ISRO Across Indian Climate Zones: A Novel Multi-Triplet Collocation Approach. Submitted to Journal of Hydrology. Preprint available on SSRN (ID: 5917277). [DOI] **(Under Review)**

Chakraborty, A., & Upadhyaya, S. (2026). Untangling Aerosol–Precipitation Susceptibility Through Thermodynamic Regimes Over the South Asian Monsoon. In 2026 AGU Annual Meeting. **Conference Paper (Submitted)**

Manuscripts in Preparation

Chakraborty, A., Upase, Y., Upadhyaya, S., Arulraj, M., & Emmanuel-Kirstetter, P. *Conditional Bias Correction of Extended Triple Collocation based merged dataset using GAMLSS*

Chakraborty, A., & Upadhyaya, S., *A Meteorology–Cloud Regime Framework for Investigating Aerosol–*

Precipitation Relationships using PCA

Chakraborty, A., & Upadhyaya, S., *A Meteorology–Cloud Regime Framework for Investigating Aerosol–Precipitation Relationships using Self Organising Maps and Explainable AI*

Chakraborty, A., & Upadhyaya, S. *Review on varying effects of aerosol type on precipitation*

Upase, Y., **Chakraborty, A.**, Arulraj, M., Mitra, K. & Upadhyaya, S., *Optimizing IMERG Bias Correction over Complex Terrain: A Comparison of Deterministic and Probabilistic Deep Learning Loss Functions in the Western Ghats*

Ponukumati, P., **Chakraborty, A.**, & Upadhyaya, S., *Rainfall Characteristics and Meteorological Controls on the Washout Efficiency of Urban $PM_{2.5}$ and PM_{10}*

Kumar, B., **Chakraborty, A.**, Upase, Y., Khadagaraj, S., Das, S. & Upadhyaya, S. *Assessment of GPM IMERG Final Run V07B Precipitation Estimates over Mumbai at Sub-daily Timescales Using the Mumbai MESONET Network*

Sarawagi, A., **Chakraborty, A.**, & Upadhyaya, S.. *Satellite-Based Rainfall Nowcasting Using Spatiotemporal U-Net: Impact of Loss Function Selection*.

Ponukumati, P., **Chakraborty, A.**, & Upadhyaya, S.. *Temporal Dynamics of Urban Particulate Matter - Insights from Comparative Wet and Dry Periods.*

Bhala, M., Golani, J., Sarawagi, A., **Chakraborty, A.**, Arulraj, M., Ramamurthy, S., Upadhyaya, S., Kurte, K. *When a Vision-Language Model Reads Disaster Rainfall Maps: How Fake Text and Social Pressure Change Its Answers*

Conferences:

Chakraborty, A., & Upadhyaya, S. (2026). Untangling aerosol–precipitation susceptibility through thermodynamic regimes over the South Asian monsoon. In 2026 AGU Annual Meeting. **Conference Abstract (Submitted)**

Chakraborty, A., & Upadhyaya, S. (2025). Disentangling the effects of aerosols on precipitation under varying meteorological conditions. In The XIIth Scientific Assembly of the International Association of Hydrological Sciences (IAHS 2025). **Poster presented**

Chakraborty, A., Upase, Y. N., & Upadhyaya, S. (2024). Extended triple collocation: A novel approach for unbiased comparison of GSMaP and GSMaP ISRO Rain using independent products SM2RAIN and ERA5. In 2024 AGU Annual Meeting. **Poster presented**

Chakraborty, A., Upase, Y., & Upadhyaya, S. (2024). Extended triple collocation for enhanced precipitation analysis: Evaluating and integrating GSMaP ISRO Rain, GSMaP, CHIRPS, and ERA5 rainfall data. In 11th International Precipitation Working Group Workshop. [DOI]. **Poster presented**

Chakraborty, A., Kumar, B., & Upadhyaya, S. (2024). Predicting and mapping flood susceptibility: Leveraging explainable AI and GIS techniques. In 2024 IEEE India Geoscience and Remote Sensing Symposium (InGARSS). **Oral presentation**

Atheeswaran, B., Jain, S., **Chakraborty, A.**, & Upadhyaya, S. (2024). Enhancing river discharge forecasting with multivariate LSTM. In 2024 IEEE India Geoscience and Remote Sensing Symposium (InGARSS). **Oral presentation**

Academic Training & Summer Schools

2026 ACM Second Asian School on High-Performance Computing and Artificial Intelligence
Association for Computing Machinery (ACM), RIKEN Center for Computational Science (R-CCS), Kobe, Japan | 30 January–3 February 2026

Training in high-performance computing, large-scale numerical simulation, AI for scientific computing, and hands-on use of the Fugaku supercomputer and SCALE-RM cloud-resolving model.

2026 ARM Big Open Data Science Summer School
U.S. Department of Energy Atmospheric Radiation Measurement (ARM) User Facility, National Weather Center, University of Oklahoma, USA | 18–22 May 2026

Training in atmospheric observations, scientific computing, AI, and open-science practices using ARM data and research infrastructure.

Awards, Grants and Fellowships:

May 2026	Travel Grant - ARM 2026 Summer School: Big Open Data Science
	[National Weather Centre, Oklahoma, USA] *(Sponsored by U.S. Department of Energy)*
Jan 2026	Travel Grant - ACM Second Asian School on High-Performance Computing and Artificial Intelligence *(Sponsored by JICA)*
	[Riken Center for Computational Science, Kobe, Japan]
Dec 2024	Best Paper Award 🏆 – AI in Natural Disaster Management, IEEE InGARSS 2024
Dec 2024	Travel Grant – IEEE InGARSS 2024
	[National Institute of Technology, Goa, India]
Sep 2023 - present	Seed Grant-funded Ph.D. Fellowship, IIT Hyderabad
Sep 2022 - Sep 2023	JRF Fellowship – Department of Space, Govt. of India
Jul 2020 - May 2022	GATE Scholarship – M.Tech in Civil Engineering

Positions of Responsibilities

Doctoral Representative, Department of Civil Engineering, IIT Hyderabad
- Representing doctoral students within departmental academic and administrative matters

American Geophysical Union (AGU), Precipitation Technical Committee
- Member of Monthly Newsletter team
- Member of ESC Precip sub-committee